Most Important Questions By Aashi Mathematics

Based on the analysis of multiple question paper sets provided in the sources, here is a curated list of **Important Questions (IMP Questions)**. These specific problem types appear frequently and carry high weightage (especially in Sections D and E).

1. TOP 5 "SURE-SHOT" LONG ANSWER QUESTIONS (5 MARKS EACH)

These questions follow a nearly identical pattern in every paper. Mastering these guarantees a significant portion of your Section D score.

- **Matrices (System of Linear Equations):**
    - Question: Solve the system of equations using the matrix method:
      $x - 2y = 10$, $2x - y - z = 8$, $-2y + z = 7$.
    - Alternative: Given $A = egin{bmatrix} 1 & 2 & -3 \ 2 & 3 & 2 \ 3 & -3 & -4 \end{bmatrix}$, find A^{-1} and use it to solve a system of equations.
    - Word Problem Variant: A furniture workshop produces chairs, tables, and beds. Determine the production units using the matrix method.
- **3D Geometry (Shortest Distance & Image):**
    - Question Type A: Find the **shortest distance** between the lines:
      $ ec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$ and
      $ ec{r} = (3\hat{i} + 3\hat{j} - 5\hat{k}) + \mu(4\hat{i} + 6\hat{j} + 12\hat{k})$.
    - Question Type B: Find the **foot of the perpendicular** drawn from point $(1, 2, 3)$ to the line $ rac{x+2}{5} = rac{y+1}{2} = rac{-z+4}{3}$, or find the **image** of a point in a line.

- **Application of Integrals (Area Under Curve):**
    - Question: Using integration, find the area of the region bounded by the ellipse $ rac{x^2}{16} + rac{y^2}{4} = 1$ between the lines $x = -2$ and $x = 2$.
    - Alternative: Sketch the graph of $y = x^2$ and find the area bounded by $y = 9$, $x = 0$, and $y = x^2$.
- **Linear Programming (LPP):**
    - Question: Solve graphically to maximize/minimize Z. Example: Maximize $Z = 50x + 70y$ subject to constraints $2x + y \geq 8$, $x + 2y \geq 10$, $x, y \geq 0$.
- **Differential Equations:**
    - Question: Solve the differential equation $ rac{dy}{dx} = \cos(x) - 2y$ (Linear form).
    - Alternative: Find the particular solution for $(x^2 - y^2)dx + 2xy dy = 0$ (Homogeneous form).

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2. IMPORTANT CASE STUDY TOPICS (4 MARKS EACH)

These topics are frequent candidates for Section E (Case Studies).

- **Probability (Bayes' Theorem & Total Probability):**
    - Scenario: A population where people are tested for a disease. You are given the probability of having the disease and the accuracy of the test. Find the probability that a person actually has the disease given a positive test result.
    - Scenario: An airplane encountering different types of turbulence (Severe, Moderate, Light) and arriving late. Calculate the probability of late arrival.
- **Relations and Functions:**
    - Scenario: A relation R is defined on a set of students (e.g., students in a railway museum or seating arrangement). Check if the relation is **Reflexive, Symmetric, and Transitive**.
    - Scenario: Check if a function defining roll numbers or positions is **One-One (Injective)** or **Onto (Surjective)**.

- **Maxima and Minima (Derivatives):**
  - Scenario: A carpenter making a wooden box with a square base and fixed volume. Determine the dimensions that **minimize the surface area** (cost of painting).
  - Scenario: Finding the rate of change of an angle of elevation (e.g., a speed camera tracking a car).

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3. FREQUENTLY REPEATED SHORT ANSWER QUESTIONS (2-3 MARKS)

- **Vectors:**
  - Find the value of λ if vectors are perpendicular ($ ec{a} \cdot ec{b} = 0$) or parallel ($ ec{a} = k ec{b}$).
  - Find a unit vector perpendicular to both $ ec{a}$ and $ ec{b}$.
- **Inverse Trigonometry:**
  - Find the domain and range of functions like $\sin^{-1}(1-x)$ or $\cos^{-1}(x^2-4)$.
- **Integrals:**
  - Evaluate definite integrals using properties, e.g., $\int_0^{\pi/2} rac{\sin x}{\sin x + \cos x} dx$ or similar forms involving $ an^{-1}$.